

Project Report

Olist E-Commerce Analysis Report

Objective

The objective of this project is to analyse the Olist e-commerce dataset to understand business performance across sales, customers, products, and delivery operations. The goal is to extract actionable insights that can help improve revenue, customer satisfaction, and operational efficiency.

About the Data

- Dataset: E-commerce (Olist)
- Total Orders: ~99K
- Total Revenue: ~15.8M
- Tables Used:
 - customers
 - orders
 - order_items
 - products
 - sellers
 - payments
 - reviews
 - geolocation
 - category_translation

The data includes information on customer demographics, order details, product categories, seller activity, payment methods, and customer reviews.

Tools & Technologies

- SQL Server
 - Data cleaning
 - Joins and relationships
 - KPI creation
 - Views for analysis
 - Power BI
 - Data modelling
 - Interactive dashboards
 - KPI visualization
 - Multi-page reporting
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Dashboard Pages Overview

1. Overview → Business KPIs and Order value by City.
 2. Sales & Growth → Sales and Growth KPIs and trends and order patterns over time.
 3. Product Analysis → Some KPIs and Category performance and Analysis.
 4. Customer Analysis → Customer behaviour and Retention Analysis.
 5. Delivery & Reviews → Delivery and Reviews KPIs and Operational efficiency and satisfaction.
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Key Insights

Sales & Revenue

- Total revenue reached ~15.8M across ~99K orders
 - Average order value is relatively low (~160), indicating a volume-driven business model
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Customer Insights

- Major revenue comes from large urban areas like São Paulo
 - A small number of high-value customers contribute significantly to total revenue
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Product Performance

- Categories like **Health & Beauty** and **Bed & Bath** generate the highest revenue
 - Demand is strongest in lifestyle and household-related products
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Delivery Performance

- ~93% of orders are delivered on time or early
 - Average delivery time is around 12 days
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Customer Satisfaction

- Early deliveries strongly correlate with higher ratings
 - However, low ratings still occur even for early deliveries, indicating other factors like product quality impact satisfaction
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Recommendations

1. Improve Customer Retention

- High-value customers are mostly one-time buyers
 - Introduce loyalty programs or targeted marketing
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2. Focus on Top Categories

- Invest more in high-performing categories
- Optimize inventory and promotions for these segments

3. Enhance Product Quality Monitoring

- Since delivery is not the only issue affecting ratings
- Focus on product quality and seller performance

4. Reduce Late Deliveries Further

- Even a small delay percentage impacts customer experience
- Improve logistics in underperforming regions

5. Expand in High-Performing Cities

- Strengthen operations and marketing in top cities
- Use geographic insights for targeted growth

Conclusion

The analysis shows that the business operates on a high-volume, low average order value model with strong delivery performance. While logistics are efficient, customer satisfaction depends on multiple factors beyond delivery speed. By focusing on customer retention, product quality, and high-performing regions, the business can significantly improve both revenue and customer experience.